

NewsGuardX: Outcome-oriented news filtering and safe headline generation

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Chapter 1

Introduction

This section dives into the core objectives of the research, laying the foundation for understanding the role of causality techniques in evaluating news reliability. It outlines the problem domain and research problem statement, offering a comprehensive understanding of the scope of the research.

1.1 Problem Domain

Understanding causal relationships in news headlines is of high importance in fields such as journalism, policy analysis, and the automated detection of misinformation. The rapid spread of digital news through online channels requires systems to be developed capable of interpreting and representing complex cause-and-effect relationships present in unstructured text. Traditional Natural Language Processing techniques, such as sentence classification models, do not have the capability to map complex causal chains and interrelations in news headlines. This limitation becomes even more evident with large-scale datasets where cause-and-effect relationships span through multiple sentences present in the news headline.

However, causal reasoning from a text is an extremely challenging problem. Most current NLP systems mainly depend on very superficial linguistic hints such as explicit conjunctions ("because," "due to") to infer causal relations. Such methods cannot capture the broader and more implicit causal dynamics usually embedded in news headlines. For instance, in large-scale data where news headlines relating to political conflicts to economic policies may occur, cause-effect links are hardly explicit and the extraction of them calls for more sophisticated yet contextually aware techniques.

The rule-based or the most complex **RoBERTa** and **BERT** type of traditional machine learning models have been applied and acquired outstanding results at the sentence level of causal classification. For instance, *Bhatia, A., et al. (2023) [1]* demonstrated that

classification of causal events in news articles is possible with the help of RoBERTA. However, the drawback to this method is that it fails to identify more than one cause-and-effect pairs within the same sentence or across the paragraphs. It often classifies whether a sentence is causal but it remains unclear as to what cause-effect relates directly to the text under consideration. Moreover, in many cases news articles contain nested causal structures within which one or several effects may emerge from a single cause, or conversely, one effect becomes the cause of another event. These inter dependencies thus far, remain largely unexplored.

Another major limitation in this field is the use of Causal Knowledge Graphs. A causal knowledge graph represents the causal relations in a much more structured and organized form, thus enabling machines to reason over them. *Gopalakrishnan, S., et al. (2023) [4]* aimed to develop such graphs from business text. Although it was a power method to synthesize knowledge from unstructured data, it was really not optimized for dynamic and evolving datasets like news headlines. To address this issue *Tan, F.A., et al (2023) [14]* aimed on developing causal knowledge graphs for specific news texts not headlines but whole articles in pieces of sentences. To bridge the gap between pattern based approaches for KG's as used by *Gopalakrishnan, S., et al. (2023) [4]*, they went for a bert based model fine tuning it alongside the pattern based approaches that were already formed. The approach had more promising results than *Gopalakrishnan, S., et al. (2023) [4]* but still they do not have a proper causal graph which can yield the information with the know how of causality (basically cause-effect-trigger) the basis or evaluation for this technique was simple recall-focused which does not employ that the model Bert is reliably without any validation whatsoever. Furthermore test set used in the dataset would not contain any temporal information for time based causal entity extraction.

Moreover, the addition of Large Language Models (**LLMs**) in causal reasoning workflows has shown mixed results. *Bhatia, A., et al. (2023) [1]* explored how large language models like GPT-4 can help people better understand cause-and-effect relationships in text by using their reasoning and interpretation skills. However, regarding reasoning over very long text spans, the reliability and consistency of identifying causal relations are often missing from LLMs. The output of the model appears highly dependent on the structure of the prompt and can significantly vary across examples. This introduces the challenge of even determining when and how LLMs correctly identify causal relations as well as how to improve for real-world applications such as news understanding. Also stated in *Kiciman, E., et al. (2023) [7]* LLMs have the capability of extracting causal relationships from any kind of texts be it news or unstructured business text but the limitation still confines to limited context window size.

1.2 Research Problem Statement

The primary research problem, which this study attempts to address, involves improving methods for *extracting, interpreting, and representing causal relationships* from large-scale news datasets with the help of detailed causal knowledge graphs. Although the current methods proposed by *Bhatia, A., et al. (2023) [1]* and *Gopalakrishnan, S., et al. (2023) [4]* have proved their abilities to detect causal relations, they fail to produce robust, large-scale causal graphs that represent multiple cause-effect pairs across an entire news article in one pass.

The recent breakthroughs in using large language models (LLMs) for causal reasoning, as presented in *Bhatia, A., et al. (2023) [1]* and *Susanti, Y., et al. (2024) [12]*, show promise to advance causal understanding. However, the lack of structured output and accuracy in these models becomes a barrier for their practical uses in real-world applications. A significant shortcoming of current approaches is their reliance on sentence-level classification that fails to capture the full complexity in the structure of news headlines. Causal relations in news frequently span across multiple sentences, paragraphs, or even articles, leading to an incomplete understanding of the relationships between events.

The method described in the paper *Causal Reasoning in Large Language Models using Causal Graph Retrieval Augmented Generation (2024) [11]* shows promise by incorporating causal graphs into the generation process using Retrieval-Augmented Generation (RAG). However, this approach has some limitations, such as a fixed depth in retrieving causal context, which restricts its ability to capture long-term cause-and-effect connections. Additionally, relying on summarization models to divide the text into smaller pieces can be challenging because poor chunking might leave out important causal links.

Our research aims to address the following key challenges:

- How can we enhance the extraction of multiple cause-and-effect relationships from short, condensed news headlines using advanced NLP techniques and Causal Graph RAG?
- How can the integration of Causal Graph RAG techniques help in resolving ambiguities when extracting causal relationships from headlines with multiple cause-effect pairs?
- How can we ensure that the cause-and-effect relationships found in news headlines are organized into a clear and easy-to-understand causal graphs?
- How can LLMs be fine-tuned to improve the extraction of interconnected causal relationships, especially in complex news headlines?

- How can we combine rule-based heuristics with LLMs to more accurately detect and verify complex causal relationships in headlines with multiple causes and effects?
- What role can a validation layer play in verifying the accuracy of LLM-generated causal relationships in news headlines?
- What challenges arise in validating multiple cause-and-effect relationships in news headlines, and how can they be overcome with a customized validation layer?

Through addressing these questions, the NewsGuardX research aims to develop a robust framework for extracting and representing causal relationships in news headlines. This framework will organize causal relations into interpretable, interconnected causal knowledge graphs, facilitating improved decision-making, policy analysis, and the safe dissemination of news content.

Chapter 2

Literature Review

This chapter takes a closer look at the literature related to causality in news. It highlights important theories and findings that lay the groundwork for our research, showing how our work connects with and builds on earlier studies in this area.

2.1 Related Research

1. Bhatia, A., et al. (2023). *MLModeler5 @ Causal News Corpus 2023: Using RoBERTa for Causal Event Classification*.[\[1\]](#)
2. Tan, F. A., et al. (2023). *The Causal News Corpus: Annotating Causal Relations in Event Sentences from News*.[\[13\]](#)
3. Tan, F. A., et al. (2023). *Constructing and Interpreting Causal Knowledge Graphs from News*.[\[14\]](#)
4. Gopalakrishnan, S., et al. (2023). *Text to Causal Knowledge Graph: A Framework to Synthesize Knowledge from Unstructured Business Texts into Causal Graphs*.[\[4\]](#)
5. Samarajeewa, C., et al. (2024) *Causal Reasoning in Large Language Models using Causal Graph Retrieval Augmented Generation*. [\[11\]](#)
6. Liu, J., et al. (2023) *Event Causality Extraction via Implicit Cause-Effect Interactions*. [\[9\]](#)
7. Sandoval M, A., et al. (2024) *The Financial Document Causality Detection Shared Task (FinCausal 2023)*. [\[10\]](#)
8. Chen, M., et al. (2023) *CHEER: Centrality-aware high-order event reasoning network for document-level event causality identification*. [\[2\]](#)

9. Hershowitz, B., et al. (2024) *Causal knowledge extraction from long text maintenance documents*. [5]

2.1.1 Causal News Corpus using RoBERTa for Causal Event Classification

2.1.1.1 Summary of the research item

The research paper *MLModeler5 @ Causal News Corpus 2023: Using RoBERTa for Causal Event Classification* by Bhatia, A., et al. [1] investigates the use of **RoBERTa**, a transformer-based model, to classify causal relationships in socio-political news texts, particularly in protest event articles. This is part of **CASE 2023's Subtask 1**: it targets the recognition of causal relations in the sentences coming from the **Causal News Corpus (CNC)**. They experimented with different variants of the dataset-eliminators of stopwords, abbreviations, and noise cleaners. They also tested the performance on RoBERTa with some of the metrics like F1-score, recall and accuracy. The best result was when stopwords were included but abbreviations were not replaced, getting an **F1-score of 0.77437**. In this aspect again, the role of context comes forward in the classification of causal events.

Therefore, the conclusion is that stops aid the identification of cause-effects because they primarily appear in prepositions or conjunctions. Although RoBERTa performed better than prior models, in the paper, it shows where the model has room for improvement and such areas include increasing token length or hyperparameter tuning. The study's primary contribution lies in demonstrating the effectiveness of transformers for event causality detection in news texts, though it also highlights the need for further optimization.

2.1.1.2 Critical analysis of the research item

The research effectively utilizes the transformer-based RoBERTa model to handle the complexity of causal event classification. By applying robust preprocessing methods (**e.g., removal of noise, handling of abbreviations**), the authors ensure that the input data is clean and contextually sound. The use of multiple datasets allows the study to analyze the influence of specific text elements, such as stopwords and abbreviations, on the model's performance. This comprehensive approach strengthens the study's credibility by showing how the inclusion of seemingly insignificant words (like stopwords) can improve the detection of causal relationships. The comparison of different versions of the dataset also provides a deeper understanding of the model's behavior.

While the study achieves promising results, its performance (**with an F1-score of 0.7475**) remains below the baseline (**0.8191**), indicating the need for more advanced tuning and

optimization. The scope is also limited to protest-related news, which affects the model’s generalizability. Causal relationships in protest news are typically clear and direct, but other news domains (such as health or finance) may require the model to handle more complex, multi-layered causal patterns. The research does not address how well RoBERTa performs with such nuances, leaving this as a potential area for future improvement. Additionally, the binary classification approach limits the model’s ability to detect multiple causes and effects within the same sentence. .

2.1.1.3 Relationship to the proposed research work

The findings of this research are directly applicable to our research *NewsGuardX*, which aims to improve the safety and reliability of news headlines by constructing causal knowledge graphs. RoBERTa’s ability to detect cause-and-effect relations can be leveraged to identify causal structures in news articles, ensuring that headlines accurately reflect underlying events without misrepresentation.

The study’s focus on including stopwords to improve how cause-and-effect relationships are detected is also important for our research, where maintaining these contextual cues can ensure more accurate headline filtering. Furthermore, the research identifies a key gap—its inability to handle multiple causes and effects in a single sentence. This gap aligns with our research of not only detecting simple causal relationships but also interpreting complex, multi-causal events, thereby enhancing the system’s capacity for more accurate news interpretation. The insights gained from this research will guide the design of our research, especially in terms of optimizing model performance and broadening the dataset to cover more diverse news categories.

2.1.2 The Causal News Corpus: Annotating Causal Relations in Event Sentences from News

2.1.2.1 Summary of the research item

The research *The Causal News Corpus: Annotating Causal Relations in Event Sentences from News* by Tan, F. A., et al. [13] presents the creation of the Causal News Corpus (CNC), a corpus of protest event news sentences annotated to classify whether they contain causal relationships. The dataset consists of 3,559 sentences in which 1,957 are causal and 1,602 are non-causal. For that gap, the authors have created the CNC which is a more diversified construction than other corpora consisting only of explicit connectives like "because" or "since". The corpus contains event causality annotations where both cause and effect arguments can remain in the form of clauses or need not be so. Hence it’s far more flexible with respect to the approaches of others like forms of the **Penn Discourse**

Treebank (PDTB) or CausalTimeBank (CTB).

Instead, the paper used a neural network built on top of a pre-trained BERT model to classify causal event sentences and scored an F1-score of **81.20%** on test set indicating that indeed the corpus allows the models to effectively identify causal relations. Moreover, the paper can prove its transferability as models trained on the **CNC** dataset were able to generalize to other datasets like **PDTB** and **CTB**. It also determined whether the use of crowdsourced annotations is effective, concluding that the crowd cannot be matched by experts in terms of accuracy; therefore, it is not an easy task that requires expertise approval.

2.1.2.2 Critical analysis of the research item

The most important strength of this research is its valuable contribution to event causality, which depicts the shortcomings prevalent in the existing causality datasets. This research approach is one that produces rich causal constructions in which cause and effect need not necessarily be in clear clauses—that is the type that is more detailed and realistic in terms of annotating real sentences of events. The results also show that using a **BERT**-based neural network goes very well in handling the task with at least small margins without much adjustment if there is ever a need. Thus, validation of the dataset by using external sources such as **PDTB** and **CTB** further establishes the trustworthiness of annotations and shows that the corpus can be used in wider applications. This crowdsourced annotation experiment exemplifies the dangers of inferring causality and therefore underscores the importance of including experts in this arduous process.

Although the paper achieves great results on the **CNC** dataset, there are a few limitations. One major issue is that the BERT-based model only identifies if a sentence has a causal relationship, but it doesn't specifically label the cause and effect parts of the sentence. This lack of detailed tagging (**like <cause> and <effect> labels**) limits its usefulness in tasks where understanding the specific cause and effect is important, such as generating news headlines. Additionally, the study didn't include a validation step, which could have made the results more reliable and reduced false positives.

Another problem is that the dataset was partly built using rule-based methods instead of being fully automated. This reliance on manual or semi-automatic rules might introduce biases or inconsistencies in the data, which could impact how well the model learns and performs. Also, when tested on other datasets like **PDTB** and **CTB**, the model's performance was much lower than on **CNC**, showing that while **CNC** works well for protest-related news, more improvements are needed for it to handle other types of news effectively.

2.1.2.3 Relationship to the proposed research work

This research paper connects well with the goals of our research, which focuses on building causal knowledge graphs to improve the accuracy and safety of news headlines. The CNC’s method of finding cause-and-effect relationships in news articles is useful. The flexible rules in the CNC, which handle different ways of showing causality, can help build better models to identify complex causal patterns in the system. Using these methods will enhance understanding of cause-and-effect in news articles, ensuring that headlines accurately match the events.

The CNC’s results on how well its models work with other datasets also provide helpful guidance, especially since our research work requires to handle different types of news, not just protest-related content. These findings can aid in choosing the right datasets and improving model performance. The study also points out the difficulty of using crowd-sourced workers for labeling cause-and-effect, emphasizing the need for expert-created datasets to ensure safe and accurate headlines.

One limitation in the research is that it doesn’t label the exact cause and effect in sentences, which is something that could be improved. Developing a model that not only finds causality but also clearly tags the cause and effect will lead to more accurate headlines. Adding a validation step and automating the dataset creation process would also help avoid some of the issues found in this research.

2.1.3 Constructing and Interpreting Causal Knowledge Graphs from News

2.1.3.1 Summary of the research item

The research paper *Constructing and Interpreting Causal Knowledge Graphs from News* by Tan, F. A., et al.[14] suggest a way to automatically find causal relationships in news articles and show them as causal knowledge graphs (KGs). This work is motivated by the increasing amount of online news, which makes it hard for people, especially in the financial sector, to keep track of all important causal events. Two main methods are used in this research work:

- Extracting causal relationships using both neural network techniques (like BERT) and pattern-based methods.
- Grouping related causes and effects together to create a more connected and understandable knowledge graph.

The main contributions of the paper include a model that effectively extracts causal relationships, merging groups of arguments to reduce the number of disconnected parts of the graph, and showing how these graphs can be useful in real-world applications. By using both BERT and pattern-based techniques, the authors aim to enhance the accuracy and completeness of their findings. Their final knowledge graph showed significant improvements in connectivity, reducing the number of disconnected parts from 15,686 to just 1. The paper also highlights various use cases where these causal knowledge graphs can help with decision-making, monitoring trends, and predicting future events.

2.1.3.2 Critical analysis of the research item

The paper introduces a creative and strong way to build causal knowledge graphs by combining two methods: a pattern-based approach and a neural network model (**BERT**). This combination is an improvement over past methods that only used patterns and often missed many cause-and-effect relationships. The research work also solved the problem of disconnected parts in the graph by adding a clustering step, which groups similar causes and effects. This makes the graph more understandable and reduces repetition.

Another key aspect of the paper is that it shows how this method can be useful in real-world industries like finance and supply chains. The Knowledge Graphs help people see how different events are connected, which can help them understand market trends and predict future events. This focus on practical applications makes the research valuable for both academics and industry professionals.

Even though the method works well, there are some limitations. One issue is that the paper relies a lot on **BERT** models, which identify causal relationships but don't clearly tag which part of a sentence is the cause and which is the effect. This means the information extracted is not as detailed as it could be. Additionally, the paper doesn't include a way to double-check the accuracy of the causal relationships it finds, which could make the results more reliable.

Another limitation is that some parts of the process, like finding patterns for the pattern-based method, still involve manual work. This can introduce bias or inconsistency. Even though this paper is focused on improving accuracy and completeness, newer techniques such as LLM-based methods are available; therefore, there is room for causal knowledge extraction at a better depth and of an improved quality.

2.1.3.3 Relationship to the proposed research work

Our research goals include the development of causal knowledge graphs to enrich news headlines for better safety and accuracy. The method as introduced in this paper fits very well into our research goals since the method appears practical and useful when

combining neural networks with pattern-based methods in search of causal relations that might form a basis for the building of similar knowledge graphs. Also, the clustering techniques applied for aggregating disparate causal nodes into a more coherent graph will be helpful in increasing the coherence along with connectivity of the graphs generated in the project.

This was a study on extraction of causal relation with high recall and precision aimed at ensuring that the information flowing in the news headlines was accurate. And so, it serves as a basis for not spreading false or misleading stories. This aside, our assignment might even include trend tracking and forecasting of what is in store and such as thing as seemingly mundane can be illustrated on paper to come up with even more contextually pertinent and credible headlines. It also creates room for further work since dense cause-effect tagging is not a characteristic in the paper. Adding more causal tagging in higher resolutions along with step-checking accuracy of the causal relationships will be helpful.

2.1.4 Text to Causal Knowledge Graph: A Framework to Synthesize Knowledge from Unstructured Business Texts into Causal Graphs

2.1.4.1 Summary of the research item

The paper titled *Text to Causal Knowledge Graph: A Framework to Synthesize Knowledge from Unstructured Business Texts into Causal Graphs* by Gopalakrishnan, S., et al. [4] proposes a framework known as **Text2CausalGraph**, which synthesizes knowledge from unstructured business texts into causal knowledge graphs. This work received a lot of attention since it concerns the extraction of causal relationships from complex business reports and proper management in a knowledge graph that facilitates decision-making processes. This is then applied on a dataset of company reports. And in such a dataset, the machine learning models like **BERT** and **SpanBERT** identify and extract the causal relationships, and then those extracted causal relationships are visualized in a graph.

In a nutshell, the paper integrates three significant contributions: an annotated dataset of cause-and-effect relationships, pipeline for automatically processing business reports into Knowledge Graphs, and classification of causal statements with regards to relevant stakeholder groups. That is, it allows for the automated detection of causal propositions in reports by decision-makers at the business analyst and auditing level through visualizations as well as being able to access useful information across multiple stakeholder groups.

2.1.4.2 Critical analysis of the research item

A technique of transforming casual knowledge graphs from unstructured business texts is proposed. Advanced machine learning models, such as **SpanBERT** and **DistilBERT**, can be used for this. Businesses generally see this system very useful in automatically detecting cause-and-effect relationships and filtering them into categories, hence making things easier to analyze, such as financial performance or corporate social responsibility. The transformer models used here include BERT, which actually enhances the accuracy of causes, effects, and triggers linking these together.

The most important finding of this methodology is relations that can be graphed, allowing users to visualize what different actions result in for businesses; hence, it makes complex business situations easier to understand and explore for the decision-maker.

However, the approach has certain limitations. The foremost one is that it relies on trigger words to identify cause-and-effect relationships. This might suggest that some relationships are missed because this relationship does not manifest with those words. Thus, it might exclude most cases of cause and effect. Finally, whereas models have been successfully applied, the system is weak with other advanced methods, which could have enhanced accuracy and depth in results.

2.1.4.3 Relationship to the proposed research work

This paper discusses some methodologies that are relevant to our objectives of research. Objectives of research include developing causal designed knowledge graphs in news headlines for the reliability and safety of news headlines. The extraction technique of causality as described in this paper may be considered to draw cause-and-effect pairs from the news articles. Using those methods, it may be applied for identifying and organizing causal relationships that impact the way headlines are generated.

It is centered on clustering and linking causal relations in a knowledge graph that would serve the objective of generating a clear, understandable representation of news events. The same techniques would produce accurate representations of causal chains in news articles and thus filter out misleading or unsafe headlines.

Some of the research gaps that can be associated with the method as proposed above are its adaptability toward the pattern-based approaches in extracting causal relations. This is sorted by more innovative machine learning models, like the large language models or other novel techniques. Such techniques will make even more complex and fragile causal relationships, thereby improving reliability for the knowledge graphs of casual relations.

2.1.5 Causal Reasoning in Large Language Models using Causal Graph Retrieval Augmented Generation.

2.1.5.1 Summary of the research item

It is subtitled such, *Causal Reasoning in Large Language Models using Causal Graph Retrieval Augmented Generation* by Samarajeewa, C., et al. [?], which discusses improving how large language models understand cause-and-effect. Although well-suited for general knowledge and logical reasoning at its worst, a large language model can hardly tell subtle causal relations—for example, what actually caused an event or what might happen in the future. The authors introduce a new strategy for this reason: The application of causality on causal graphs helps the models. These are sorts of diagrams that depict cause-and-effect relationships. Such information, which is given to the models as an auxiliary entity they can retrieve and apply for inference, leads to improved choices.

It works on two segments: breaking down the text into little pieces and then a summarization model feeds off this structure to identify causal relationships between events. These relationships are then captured as a causal graph where an event is regarded as a node which owns other events portraying causes and effects. The graph is then further stored as a vector database so that it can be reused later on. While answering a question, the model fetches the appropriate parts of the causal graph from the vector database. Then, those cause-effect relationships could be exploited by the model itself to answer in a more valid and logically sound manner. To test their design, authors employ two datasets: **CosmOsQA** and **AdversarialQA**, relying on common sense as well as factual reasoning. Results are over those of traditional methods concerning improving the relevance and grounding related to the answer returned by the model.

2.1.5.2 Critical analysis of the research item

This introduces an innovative solution towards the very well-known gap in large language models (**LLMs**)—causal reasoning. This makes it possible by incorporating causal graphs as an external resource that supports retrieval-Augmented generation (**RAG**) that proved to be a giant leap in handling complicated reasoning tasks. Information is structured in causally coherent manners, which help LLMs in answering questions that prove not only to be contextually relevant but also accurate in terms of cause and effect determination. This is particularly useful in applications like automated decision-making systems, chat bots, or content generation, for which knowledge of causal relationships has direct implications on the reliability of the result.

The paper nevertheless has limits. The technique will break down if the summarizer of causal relationships, or rather how well the model of summarization tracks these rela-

tionships, develops incorrectly. If subtle or complex causal links slip the summarizer, the system suffers. This becomes an even bigger concern in the case of longer or more complex text, for a summary model will probably miss some vital or important causal chains. The approach has yet another weakness, associated with the depth of fetching causal context during inference; in other words, the model isn't well adapted to varying degrees of complexity about causality differently based on a query. It would give a much more vibrant retrieval mechanism, capable of changing the depth based on the complexity of the query.

2.1.5.3 Relationship to the proposed research work

The paper falls in line with the objectives of our research NewsGuardX, which relate to improving the transparency and reliability of news headlines through causal reasoning based on factual data. A very good foundational framework in which improvements may be made to the accuracy and clarity of cause-effect detection may be looked upon in the dynamic construction of causal graphs for causality monitoring and verification behind events in news headlines. The paper would present a causal graph-based retrieval system which would be used directly in building a system to prevent headlines from becoming either misleading or untenable.

One of the key contributions from the research work is a method about organizing causally relevant information and retrieving it. It has been possible through representations of cause-effect relationships within texts graphed causally to safely allow news headlines which are contextually accurate to be generated.

This type of retrieval looks for causally aligned information and not that based on pure semantic closeness, enhancing factual accuracy, and ensuring the causal relations between events remain logical and verifiable. It is indeed powerful in the effort against misinformation and focuses well on causal transparency. The fixed-depth retrieval system developed in the paper can easily be extended to cover use in applications where the depth of causal relationships depends upon the subject matter. Then a more general retrieval system would then have to be constructed to handle a greater variety of information across multiple and different causal levels of complexity. Applications of these ideas might help develop such a system, ensuring that headlines for news stories were both factually correct and causally coherent with events in the world.

2.1.6 Event Causality Extraction via Implicit Cause-Effect Interactions

2.1.6.1 Summary of the research item

The research paper, *Event Causality Extraction via Implicit Cause-Effect Interactions* by Liu, J., et al.[9], focuses on extracting cause-effect pairs from text to improve models' ability to understand event causality. This is known as **Event Causality Extraction (ECE)**, it requires advanced reasoning to recognize implicit interactions between cause and effect events. The paper introduces an **Implicit Cause-Effect Interaction (ICE)** framework, which identifies cause and effect relationships by using templates for conditional generation. It helps the system predict causes or effects based on context, even when they aren't explicitly mentioned. The model uses additional information (**i.e event types and arguments**) during training to improve its reasoning skills. A major innovation in this paper is the use of *knowledge distillation technique*, where two teacher models transfer their knowledge to a student model. This allows the student model to perform well even when the additional information is not available during testing.

The **ICE** framework also introduces a new alignment strategy called **Cause-Effect Optimal Transport (CEOT)**, which enhances the relationship between cause and effect events. This ensures that model captures both intra- and inter- event interactions, improving the extractions of event structures. Testing on the *ECE-CCKS dataset* showed that the **ICE** framework outperformed previous methods, achieving an **8.39%** increase in F1-score. This improvement demonstrates the model's ability to identify implicit causal relationships, making it effective for tasks like machine reading comprehension and predicting future events. The study successfully overcomes limitations in existing **ECE** models that rely mostly on classification methods. By switching to a generation-based approach and adding mechanisms to use additional information, the **ICE** framework improves the models ability to understand causality.

2.1.6.2 Critical analysis of the research item

The paper offers significant contributions to the field of causal reasoning in natural language processing. The **ICE** framework moves beyond traditional classification approaches, which often fail to capture complex cause-effect relationships. By introducing a template-based generative approach and leveraging additional information during training, the authors ensure that the model develops a deep understanding of both intra- and inter- causal relationships. This methodology increases the model's capability to extract causal relationships more accurately, leading to state-of the art performance on benchmark datasets.

However, the reliance on additional information (**i.e event types and arguments**) is ef-

fective but faces challenges during deployment because this information might not always be available. Although the knowledge distillation mechanism helps reduce the problem by transferring knowledge from teacher models to students model, it adds extra complexity and increases computational costs, especially during training. Additionally, while the **CEOT** strategy is innovative, it may need further improvements to work consistently across different datasets, particularly those with more complex or varied causal structures. The computational load from using multiple teacher models also raises concerns about scaling for large real-world applications. Despite these limitations, the paper marks a significant step forward, especially by emphasizing the importance of implicit cause-effect interactions and the use of additional information during model training.

2.1.6.3 Relationship to the proposed research work

The methods in this paper fit well with the goal of our research NewsGuardX, creating reliable news headlines based on accurate cause-and-effect reasoning. The **ICE** framework focuses on finding hidden cause-and-effect connections, which can be used to make sure news headlines reflect the true reasons behind events. Using additional information during training can help build stronger links between events, leading to more trustworthy headlines.

The **ICE** framework’s method of passing knowledge from a teacher model to a student model can also help improve understanding of cause-and-effect when full information isn’t available. This is useful in news settings, where not all details about an event may be known. By using the **CEOT** strategy, the system could make sure that events in the news are logically connected, helping to maintain both accuracy and flow in the stories being told. The paper also highlights the success of using templates to extract cause-and-effect relationships, that can be used to ensure headlines are reliable.

2.1.7 The Financial Document Causality Detection Shared Task (FinCausal 2023)

2.1.7.1 Summary of the research item

The research paper titled *The Financial Document Causality Detection Shared Task (FinCausal 2023)* by Sandoval M, A., et al.[10] presents a shared task designed to identify causality in financial documents. The FinCausal 2023 task is a continuation of previous editions held since 2020, aimed at improving the detection of cause-effect relationships in financial texts. This year’s edition of the task introduced two subtasks: one in English and one in Spanish, marking the first time a non-English subtask was included. The main goal was to decide whether an event or series of events could be seen as the cause of some-

thing that happened earlier in financial documents. Participants had to label sentences with clear cause-and-effect relationships, identifying both the cause and the effect, and keeping them separate in context.

Participants were tasked with applying various methods, ranging from simple pattern matching to advanced deep learning models. For the English subtask, the dataset came from financial news articles, while for the Spanish subtask, it came from financial annual reports. Both datasets were carefully annotated by expert linguists to ensure accuracy. The paper mentions five teams participated in the English subtask and three in the Spanish one, using different models. These included transformer-based models like **RoBERTa** and **prompt-based models using GPT**. The results indicated that models designed with simpler, more traditional transformer architectures outperformed those using complex generative models.

The goal of this shared task was to help improve automated analysis of financial documents, particularly in detecting cause-and-effect relationships, which are important for decision-making in finance. The availability of annotated datasets in both English and Spanish also provides valuable resources for future research in financial natural language processing (NLP), showing how causality is expressed differently across languages and types of documents.

2.1.7.2 Critical analysis of the research item

The FinCausal 2023 task makes an important contribution to the field of natural language processing(NLP) by focusing on detecting causality in financial documents. A key strength of this research is the expansion into bilingual causality detection, which adds both diversity and depth to the challenge. By introducing a Spanish subtask for the first time, the organizers effectively broadened the scope of the task, recognizing the need for financial NLP tools that work across multiple languages. The use of annotated datasets in both the English and Spanish subtasks also helps ensure that the results are reliable and robust. The focus on causality is particularly important, as this is a complex and crucial concept for financial analysis and decision-making.

Despite its strengths, the paper points out a few limitations. While the task deals with causality, it did not fully explore more complex aspects like multi-causal relationships or chained causality. The Spanish subtask, in particular, focused only on simpler causal structures, which limits the depth of analysis and leaves room for further exploration in future editions. Another issue is that more advanced models, like **GPT-based generative models**, performed worse than simpler transformer models like **RoBERTa**. This shows that even though advanced models may bring new ideas, they may not yet be fully fine-tuned for tasks like causality detection in financial texts, which need precise, domain-specific adjustments. Another area for improvement could be the dataset itself. Financial

texts are often complex, containing implicit and indirect causal relationships. A more detailed approach to building the dataset, allowing for multi-layered or indirect causes, could make the task more challenging and provide better benchmarks.

2.1.7.3 Relationship to the proposed research work

The insights from this research are relevant to our research work NewsGuardX, of generating reliable, fact-based news headlines through causal reasoning. Both FinCausal and this research focus on understanding causality in text, making the methodologies used in the shared task highly relevant for improving causal inference in the news domain. For instance, the use of annotated datasets for detecting causality in financial texts could inspire similar approaches for training models to identify and generate causally coherent news headlines. The transformer-based architectures used in FinCausal, like RoBERTa, could also be adapted to optimize causal detection in news.

In addition, the inclusion of both English and Spanish subtasks in FinCausal aligns with the potential need for multilingual support. News headlines are often published in multiple languages, and having a system capable of detecting causality across different languages is crucial. By using bilingual datasets and adopting methods similar to those in FinCausal, the system could expand its capability to generate accurate causal narratives in multiple languages. This cross-linguistic approach ensures that reliability and accuracy are maintained across various news landscapes.

The findings from FinCausal also suggest that traditional models might perform better than more complex architectures for specific tasks. This is an important consideration, as it shows that a balance between model complexity and task specificity is necessary. By learning from the FinCausal results, the focus can be on models that offer precision in causal reasoning without compromising efficiency or scalability.

2.1.8 CHEER: Centrality-aware high-order event reasoning network for document-level event causality identification

2.1.8.1 Summary of the research item

The paper titled *CHEER: Centrality-aware High-order Event Reasoning Network for Document-level Event Causality Identification* by Chen et al.[2] focuses on improving how we identify cause-and-effect relationships between events in documents. This task, called **Document-level Event Causality Identification (DECI)**, is difficult because events are spread out across the document, and there aren't always clear words linking causes and effects between sentences.

The authors introduce a model called **CHEER (Centrality-aware High-order Event Reasoning network)**, which uses graph neural networks (GNNs) to reason about these events. CHEER builds an **Event Interaction Graph (EIG)**, showing how events and event pairs interact. A key aspect in CHEER is that it factors in the centrality of events during its reasoning, treating central events as key anchors to understand cause-effect relationships in the document. By labeling central events in datasets (**such as EventStoryLine**), the authors created features that capture the role and importance of these events. CHEER performs better than existing models on two benchmark datasets, showing that it is more effective in identifying causal relationships in documents.

2.1.8.2 Critical analysis of the research item

The **CHEER** framework presents an innovative approach to addressing the limitations of traditional **DECI models** by considering event centrality. Previous models such as **RichGCN** and **ERGO** focused on constructing document-level graphs for cross-sentence reasoning but failed to incorporate the hierarchical importance of certain events. CHEER’s ability to integrate both single-event nodes and event-pair nodes in its Event Interaction Graph (**EIG**) enables it to perform high-order reasoning, capturing transitive causal relationships. This approach aligns well with scenarios where certain central events influence most other events in a document.

The research does have certain limitations. While the integrating event centrality enhances model performance, the prediction of central events is not entirely accurate. The authors acknowledge that the centrality prediction can be further improved using more sophisticated encoding methods. Moreover, the reliance on manually annotated central events raises concerns about scalability for larger datasets. In future work, automating this annotation process could handle this issue. Despite these limitations, CHEER introduces a valuable improvement over existing methods by systematically addressing the lack of structured causal information at the document level.

2.1.8.3 Relationship to the proposed research work

The CHEER model focuses on understanding how events are connected and which events are the most important. This focus is very relevant to our research NewsGuardX, which aims to understand complicated causal relationships in news. In this context, being able to create causal graphs and check the reasoning behind event sequences is essential for making news headlines more reliable and transparent. The centrality-aware framework in CHEER can be adapted to find key events in news articles, which would help in verifying causal claims more effectively.

Additionally, CHEER’s use of Graph Neural Networks (**GNNs**) for complex reasoning

aligns with the proposed research work, which aims to create causal knowledge graphs to monitor and ensure the quality of generated headlines. By including event centrality, the system could highlight important events and make sure they are correctly linked to relevant causes and outcomes. The model's approach to multi-task learning, which combines predicting centrality with detecting causality, provides a strong base for developing more trustworthy causal reasoning systems for news applications.

2.1.9 Causal knowledge extraction from long text maintenance documents

2.1.9.1 Summary of the research item

Causal Knowledge Extraction from Long Text Maintenance Documents" deals with the extraction of causal knowledge from **Work Request Notification (WRN)** records, which is brought into practice on the manufacturing lines of nearly every industry for tracking cases of equipment failure and maintenance-related activities. Generally, such high volumes of unstructured text occurring in these records are very tough to analyze and interpret at human levels. The authors of this paper discuss a two-stage deep learning pipeline with the specific aim of extracting meaningful cause-and-effect information from these WRNs.

The first stage involves sentence-level noise removal, filtering unwanted information. This stage is of significance for WRNs because they typically contain information not relevant to the causal relations - just administrative notes or nonspecific observations. The second stage employs a transformer-based model so that entities and their relationships can be extracted jointly. In fact, it most often emphasizes causal triples consisting of the cause, effect, and their relation. A **Causal Graph Database** is constructed and then queried by the technical staff, identifying patterns of equipment failure and supporting appropriate asset-management decisions. The whole evaluation is done over the dataset of 98,000 WRN records; with achieving high **F1 score (83% and 92% for cause and effect respectively)** for both entity recognition and relation extraction, the system's success on large-scale real-world data is justified.

2.1.9.2 Critical analysis of the research item

Research has successfully overcome the challenge of extracting causal knowledge from long, unstructured texts in industrial maintenance records. Novelty to the approach of sentence-level noise removal is a strength in itself here because it tends to simplify text data for better information extraction. Not surprisingly, applying deep learning models and using **BERT** to classify noise and **SpERT** for entity-relation extraction points toward

the applicability of latest NLP techniques to such a narrow yet technological area. The study emphasizes the role of pre-processing in other areas, like jargon, abbreviations, and spelling inconsistency, and all these are relevant to making the credibility of causal information extracted. It makes it possible to construct a causal graph database on which users can query and picture causality between failures of equipment and their consequences, adding practicality to the focus.

There is an important limitation. The quality of the noise removal stage depends heavily on the existence of a corresponding annotated training dataset, which requires much expert effort. This also then would also pose challenges to scaling the approach to other domains or maintenance environments where domain experts are not available. While impressive results are obtained in the intra-sentential causal relations extraction, the inter-sentential causality may be more deserving of more study. Further limitation lies in being dependent on exact matching in the graph database, which may fail to generalize over similar but lexically varied failure descriptions. It must be mentioned that the system exhibits very high performance on this specific industrial context; however, open questions arise in portability to domains which include unstructured data.

2.1.9.3 Relationship to the proposed research work

The results of this work directly reflect the goals of NewsGuardX, especially the extraction of causal relations from unstructured text. Extraction of causal relations from headline news and interesting results can be obtained on such a two-stage pipeline introduced in this work, in particular the step of noise removal, which is very useful for NewsGuardX when it will filter out content irrelevant to complex news articles and long texts. That way, meaningful causeeffect relationships would be maintained and preserved—only those—properly appropriate for the senses of causal reasoning. A transformer-based model also for the joint extraction of entities and relationships would well fit within the aims of NewsGuardX: improving the detection of causal relationships in headlines, as well as ensuring correct tags on the ‘<cause>’, ‘<effect>’, and ‘<trigger>’ elements.

One direction further research into this area could pursue is to provide suggestions on how the process of causal graph validation for NewsGuardX presented as handling and shortcomings based on exact matching may be improved. This can be helpful in fine-tuning the system in terms of variation in language in processing news. The challenge further proved the deep learning models in the ability to extract causality from noisier, unstructured data to inform development toward the creation of more rugged causal inference systems for NewsGuardX. Applying the same techniques of noise removal and entity-relation extraction would be to the advantage of a more advanced and reliable causal reasoning layer that enhances outcome-oriented news filtering and safe headline generation for NewsGuardX.

2.2 Analysis Summary of research items

RESEARCH ITEM	SHORT SUMMARY	CRITICAL ANALYSIS	RELATIONSHIP TO PROPOSED RESEARCH
Bhatia et al. (2023) [1]	Bhatia et al., 2023, utilized RoBERTa pre-training, especially with protest-related articles, for the classification of the causal relationship in socio-political news. The best F1-score achieved is 0.77437 for the stopwords.	The algorithm is reasonably strong concerning preprocessing methods and dataset variations. It fares below the baseline performance. It can only perform binary classification and, more importantly, protest news, and its optimization needs improvement.	The findings inform causal detection in NewsGuardX. The inclusion of stopwords for better cause-effect detection can enhance headline filtering. Addresses a gap in handling complex, multi-causal events, relevant to NewsGuardX's goals.
Tan, F. A., et al. (2023) [13]	Tan et al. (2023) created a 3,559-sentence corpus for classifying causal relations in protest news. BERT-based model achieved an 81.20 F1-score.	Innovative in handling diverse causal constructions. BERT is effective but lacks detailed tagging of cause/effect. Partially rule-based creation introduces potential biases.	Useful for NewsGuardX's causal graph building. Flexible rules help identify complex causal patterns, though specific tagging and validation are areas for improvement.
Tan, F. A., et al. (2023) [14]	Tan et al. (2023) proposed using BERT and pattern-based methods to build causal knowledge graphs, improving graph connectivity significantly.	Strong combination of pattern-based and neural models. Reduces graph disconnections but lacks detailed cause/effect tagging. Manual steps could introduce biases.	The dual-method approach helps enhance causal graph coherence, which aligns with NewsGuardX's goals of accurate headline generation and trend analysis. Improvements in tagging and validation are areas for enhancement.
Samarajeewa, C., et al (2024) [11]	This research integrates RAG with large language models to enhance causal reasoning by using pre-built causal graphs to retrieve relevant information for tasks like question answering and summarization.	Combining causal graphs with language models improves the ability to reason causally, but it relies on the quality of pre-built graphs, limiting scalability and performance in complex or conflicting scenarios.	Highly relevant for NewsGuardX. GraphRAG can enhance causal reasoning in news headline generation, improving safety and reliability. Automating graph creation or making them dynamic can address current limitations.

Table 2.1: Analytical Summary (1)

RESEARCH ITEM	SHORT SUMMARY	CRITICAL ANALYSIS	RELATIONSHIP TO PROPOSED RESEARCH
Gopalakrishnan, S., et al. (2023) [4]	The Text2CausalGraph framework, applied to business texts, extracts causal relationships using SpanBERT and BERT, organizing them into a stakeholder-driven knowledge graph.	The system effectively organizes complex business information into causal graphs, aiding decision-making. However, it relies on specific trigger words and lacks more advanced methods like GraphRAG.	Provides a useful framework for NewsGuardX to extract and visualize causal relationships in news. The method's clustering and linking techniques can help improve the coherence of causal chains in headlines. Adding advanced models like GraphRAG can address its current limitations.
Liu, J., et al. (2023) [9]	The framework of ICE strives to obtain implicit cause-effect pairs from complex text by producing knowledge transfer among multiple models. In this regard, it is aimed to achieve better identification of hidden causal links within sentences and diversify the contexts of correct causality extraction.	It has demonstrated significant improvement in detecting implicit causal relations but still suffers in cases of highly ambiguous or context-dependent causality. What it can be proud of is the transfer learning mechanism within, which would enable it to perform more dominantly across domains.	This technique can be applied in NewsGuardX in order to increase the detection of causal implicit relations in news headlines especially where multiple causes are involved by enhancing the accuracy of causal tagging.
Sandoval M, A., et al. (2024) [10]	FinCausal is a shared task for the identification of cause-effect relations in financial texts. FinCausal uses models like RoBERTa for explicit causal link detection in financial documents. Experiments have used the English and Spanish datasets, actually underlining the multilingual approach to processing financial text.	More complex architectures fail to align performance with RoBERTa, and even simpler ones do, so model complexity is not very highly correlated with accuracy. It rather exposed the role of precision and understanding the context associated with detecting causality in finance.	This study, therefore, demonstrates how much simpler models like RoBERTa may work incredibly well on NewsGuardX, particularly headlines that might be a hot spot of need for clear cause-effect tagging without over-complication, but still there is room for a more better model like Llama, mistral, etc, which outperform the smaller models.

Table 2.2: Analytical Summary (2)

RESEARCH ITEM	SHORT SUMMARY	CRITICAL ANALYSIS	RELATIONSHIP TO PROPOSED RESEARCH
Chen, M., et al. (2023) [2]	CHEER stands for An event reasoning model with centrality in events within causality detection, and then cross-document analysis holds to build up relationships and causal effects between the events to enhance reasoning in multi-events.	The model is good at finding event centrality across documents, but its effectiveness degrades on very fine-grained or sentence-level causality. It performs very well on multi-document and multi-event causality but lacks focus on finer details in single sentences.	This is related to NewsGuardX which, indeed performs cross-document analysis of news stories. Determination of important central events along with reasons for their occurrence can enhance accuracy of generated causal graphs.
Hershowitz, B., et al. (2024) [5]	This paper attempts to extract causal knowledge from the Work Request Notification (WRN) records, capturing equipment failures and activities of maintenance. Here, the task proposes a two-stage pipeline in deep learning: stage one eliminates unwanted noise present in sentences while stage two uses a transformer-based model for the purpose of entity and relationship extraction toward the formation of causal triples. The cause and effect recognition of the system yields a high F1 score of 83% and 92% respectively, thereby testifying the robustness on a large dataset of 98,000 WRNs.	It does well with long, unstructured text because of noise removal and deep learning, especially with employment of BERT and SpERT for improved causal extraction. Still, it will depend upon a good quality annotated training dataset, and thus does not scale well into other domains. Inter-sentential causality is under-explored and was not strong in this, but its reliance on exact matching would hurt broader applicability across different contexts; intra-sentential relations are fine, though.	The findings closely match the objectives of NewsGuardX, focusing on extracting causal relations from unstructured news headlines. The proposed noise removal and transformer-based entity extraction can improve the detection of causal links to ensure proper tagging of cause, effect, and trigger entities. Future work could be focused on enriching the validation of causal graphs to incorporate language variations in the processing of news and to increase system strength in causal reasoning for better news filtering and headline generation.

Table 2.3: Analytical Summary (3)

Chapter 3

Proposed Approach

3.1 Overview:

NewsGuardX is designed for ensuring that safe and accurate headlines are to be generated by combining causal reasoning, LLMs and graph-based validation.

The system mainly focuses on two important factors:

- Identification of headline single/multiple cause-effect-trigger relations.
- That solely authentic, safe, and correct headlines are produced.

News Headlines that will be generated safely and accurately to be valid. The approach that we now advance is the current research, which assists in improving our understanding causality in NLP and aligning with the current best practices up to 2023. onwards.

3.2 Design Architecture with Stepwise Operational Guide:

3.2.1 Step 1: Data Preparation and Data Cleaning

- The first stage involves cleaning raw news data/headlines gathered from multiple sources, including the Causalnews Corpus created by Tan, F., et al (2023) [13]. GPT-prompted tagged sentences are combined with manually annotated headlines. This stage removes superfluous noise and tags accurate sentences to create an organized dataset.

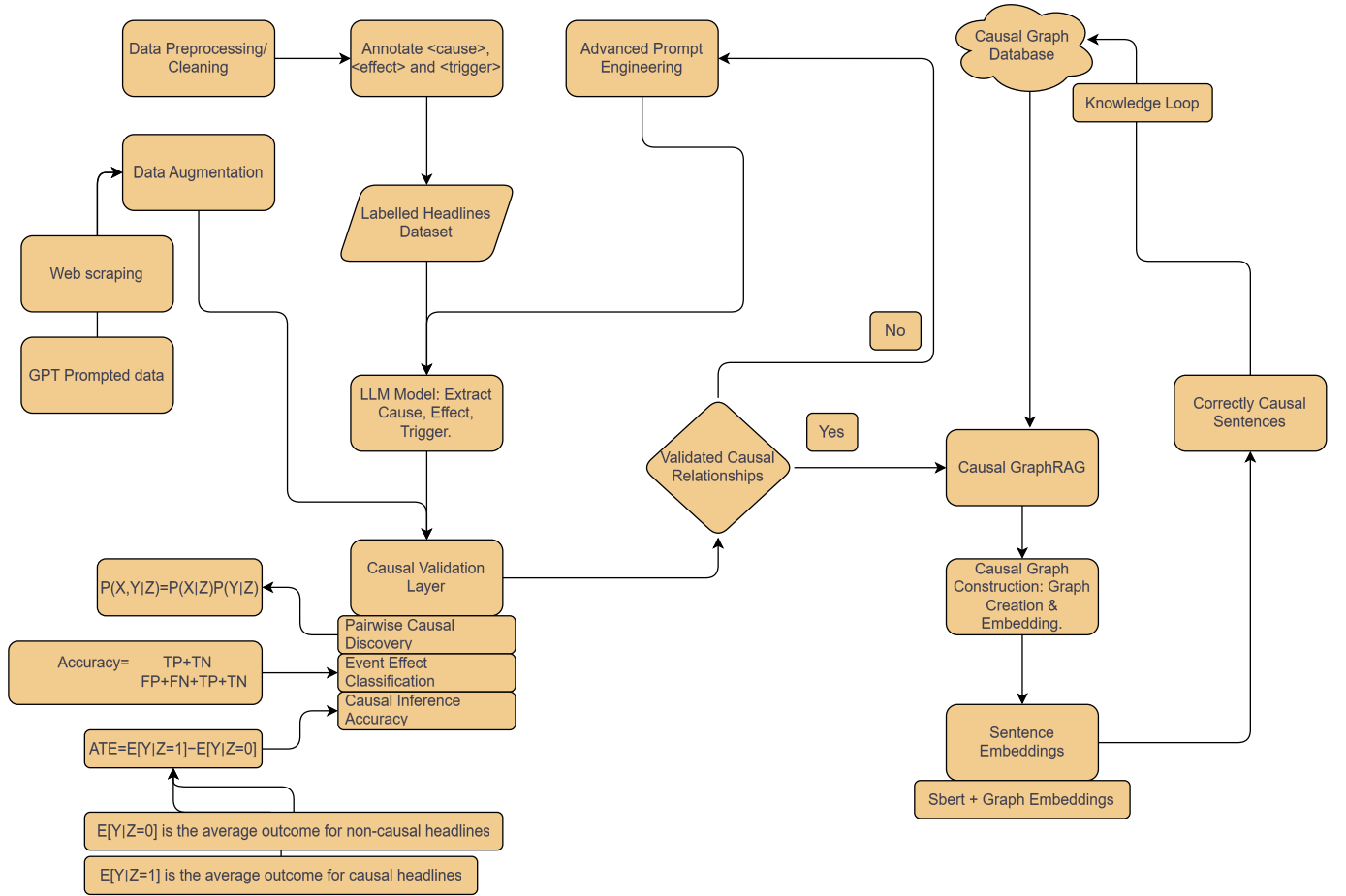


Figure 3.1: NewsGuardX Architecture

- **Why this Step?** Clean data and annotated data is required for accurate causal event extraction.
- **Reference:** Hershowitz B. et al. (2024) [5] for maintaining clean text for causal knowledge extraction.

3.2.2 Step 2: Annotate <cause>, <effect>, and <trigger>

- At this stage, causal events in the cleaned dataset are labeled with tags such as <cause>, <effect>, and <trigger>. This annotation prepares the dataset for training the LLM to identify causal relationships.
- **Why this Step?** Clear annotation is crucial for training the model to understand cause-and-effect relationships inside phrases.
- **Reference:** Bhatia A. et al. (2023) [1] for using annotated data in causal event classification.

3.2.3 Step 3: Data Augmentation

- Augmentation approaches, such as creating new training instances from GPT-prompted data, are used to enhance the number of training data points. This improves the labeled dataset, particularly in low-resource contexts.
- **Why this Step?** Data augmentation improves model generalization by producing a more diverse set of training instances.
- **Reference:** Samarajeewa C. et al. (2024) [11] for data augmentation in causal event analysis

3.2.4 Step 4: Advanced Prompt Engineering

- This stage entails utilizing personalized prompts to extract specific causal links from the LLM. It allows the model to focus on identifying causal triggers and connecting them to cause-effect pairs.
- **Why this Step?** Prompt engineering guarantees that the LLM is correctly guided to derive causal information from news texts.
- **Reference:** Wei J. et al. (2022) [15] for using chain-of-thought prompting in large language models for causality.

3.2.5 Step 5: LLM Model for Causal Extraction

- Labeled data is loaded into an LLM like Llama or mistral to extract causes, effects, and triggers from phrases. This is a critical stage in which the model identifies causal events and their relationships.
- **Why this Step?** LLMs excel in understanding context and inferring major causal relationships from unstructured text.
- **Reference:** Samarajeewa, C., et al. (2024) [11] for using causal graphs RAG with LLM.

3.2.6 Step 6: Validation Layer

- This stage determines the correctness and validity of the model's extracted causal linkages. Additional tests and cross-referencing against a causal knowledge base may be part of the validation process.
- **Why this Step?** Before proceeding with any further processing, validation ensures that the detected causal links are correct and reliable.
- **Reference:** Kiciman E. et al. (2023) [7] for discussing important validation strategies.

3.2.7 Step 7: Causal Validation Layer Insides

- This layer validates causal linkages using approaches such as **Pairwise Causal Discovery (PCD), Event Effect Classification, and Causal Inference Accuracy**. This phase ensures that the detected causal relationships represent true interdependence rather than spurious correlations.
- **Reference:** Lu C. et al. (2022) [3] and Lin C. M. et al. (2024) [8] for causal learning algorithms and validation.

3.2.8 Step 8: Causal GraphRAG (Retrieval-Augmented Generation)

- If the causal linkages are not confirmed, the system use Causal GraphRAG to extract important causal facts from a knowledge source and enhance the resulting words. This level seeks to improve headline development by leveraging an external knowledge source, such as the Neo4J graph database. The knowledge repository already has cause-and-effect trigger-tagged sentences that signal significance.

- **Why this Step?** Causal GraphRAG enables the system to merge causal graph retrieval and production, hence increasing the overall quality of generated headlines.
- **Reference:** Samarajeewa C. et al. (2024) [11] used causal RAG and have promising results therefore will use causal graphRAG for more further improvements.

3.2.9 Step 9: Causal Graph Construction

- Validated relationships are used to build a causal network. The network consists of nodes (events) and edges (causal links). Use graph embeddings (Node2Vec or TransE) to generate a causal graph. These embeddings will represent the connections between causes, effects, and triggers as graph nodes.
- **Why this Step?** Graph-based embeddings accurately depict the structure and interactions between nodes (events).
- **Reference:** Gopalakrishnan S. et al. (2023) [4] for constructing causal knowledge graphs but causal graphs have better visualization and understanding of causality.

3.2.10 Step 10: Synergizing Sentence and Graph Embeddings

- SBERT generates sentence embeddings in a low-dimensional space. This stage aids in capturing the semantic meaning of statements, enabling for proper graph querying and searching for accurate responses in the Neo4J graph database.
- **Why this step?** SBERT embeddings give a compact and semantically rich representation that can aid in comprehending sentence relationships.
- **Reference:** Hershowitz et al. (2024) [5] discusses sentence embeddings for causal knowledge extraction, where SBERT provides semantic embedding capabilities.

3.2.11 Final Step: Dynamic Safe Headline Generation via Active Learning Loop

- After validation, add validated phrases with accurate causal tags to a Causal Knowledge Base (stored in **Neo4j**) and continue to add when complicated multiple cause linkages occur more frequently. This will improve the capability of LLM. We also added another stage for safe headline generation: after validation, we just give certain guidelines to the LLM to generate a safe headline from the original one, producing accurate news headlines with safety terms added.

- **Why this Step?** Continuously upgrading the knowledge base guarantees that the model learns from fresh validated complex multiple causal linkages, hence improving its accuracy over time.
- **Reference:** Hobbhahn et al. (2022) [6] explains how LLMs can be improved by integrating them with knowledge bases for better causal understanding. In our case we have our graph database.

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